CSC301: Fundamentals of Programming Languages

Syllabus – Fall 2026

Time: Section 1 MWF 10-10:50, Location: Tyler Hall Room 108

Webpage: https://lutzhamel.github.io/CSC301 or BrightSpace
Prerequisites: CSC212

Instructor:
Prof. Lutz Hamel
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office: Tyler 251

Course Description

Language enables thought. In this course we study a class of formal languages known as programming languages. Like natural languages, these formal languages enable us to reason about algorithms and procedures to solve computational problems on computers. However, their formal nature restricts the kind of meanings particular language constructs can assume and therefore makes them amenable for the execution on a computer.

Over the years many different programming language dialects have evolved to address particular technical issues, e.g., object-oriented languages, real-time languages, database query languages, logic languages, etc. Here we study the major structures of modern programming languages. Understanding not only the syntax of a language but also the semantics and implementation techniques of this language will allow you to design better programs. Having deeper insights into the design of a programming language will also enable you to learn new programming languages much faster. Having a thorough understanding of today's languages allows you to design the programming languages of tomorrow.

Objective

Upon completion of this course

- You will be able to discern and contrast the major programming language paradigms in use today.
- You will be able to pick an appropriate language for the job at hand.
- You will have deeper insight into the evolution of programming languages.

Required Text

Modern Programming Languages: A Practical Introduction, Adam Brooks Webber, Franklin, Beedle & Associates, Any Edition.

Software

Throughout this course we will be using various programming language and software development environments including Asteroid and Prolog. More details will be given on the course website.

Grading

Assignments	20%
Midterm	40%
Final	40%

Grading Key

Symbol*	Start %*
F	0
D	60
D+	67
C-	70
C	73
C+	77
B-	80
B	83
B+	87
A-	90
A	93

Policies

- Check the website (often)! I will try to keep the website as up-to-date as possible.

- **Promptness, participation,** and **adequate preparation** for each class are expected. If you are absent, it is your responsibility to find out what you missed (e.g. handouts, announcements, assignments, new material, etc.)
- **Make-up exams** will **not** be given without a valid excuse, such as illness. If you are unable to complete a scheduled examination due to valid reasons, please inform myself, or the department office in Tyler Hall, prior to the exam time. Under such circumstances, you are not to discuss the make-up exam with any other class member until after a make-up exam has been completed.
- **Software piracy** will be dealt with exactly like stealing of university or departmental property. Any abuse of computer or software equipment will subject to disciplinary action.
- Any student with a documented disability should contact me early in the semester so that we can make reasonable accommodations to support your success in this course. You should also contact Disability Services for Students, Office of Student Life, 330 Memorial Union, 874-2098

***Academic Enhancement Center*:** Located in Roosevelt Hall, the AEC offers free face-to-face and web-based services to undergraduate students seeking academic support. Peer tutoring is available for STEM- related courses by appointment online and in-person. The Writing Center offers peer tutoring focused on supporting undergraduate writers at any stage of a writing assignment. The UCS160 course and academic skills consultations offer students strategies and activities aimed at improving their studying and test-taking skills. Complete details about each of these programs, up-to-date schedules, contact information and self-service study resources are all available on the AEC website, uri.edu/aec.

Tentative Schedule

* Why study programming languages?
* Meet our languages: Asteroid and Prolog
* Asteroid – The Imperative Basics
* Types and Type Systems
* Exploring more of Asteroid's Types
* Functional Programming with Asteroid
* Recursion
* Memory management
* Polymorphism
* Logic Programming with Prolog
* Language specification & Implementation
* Formal semantics